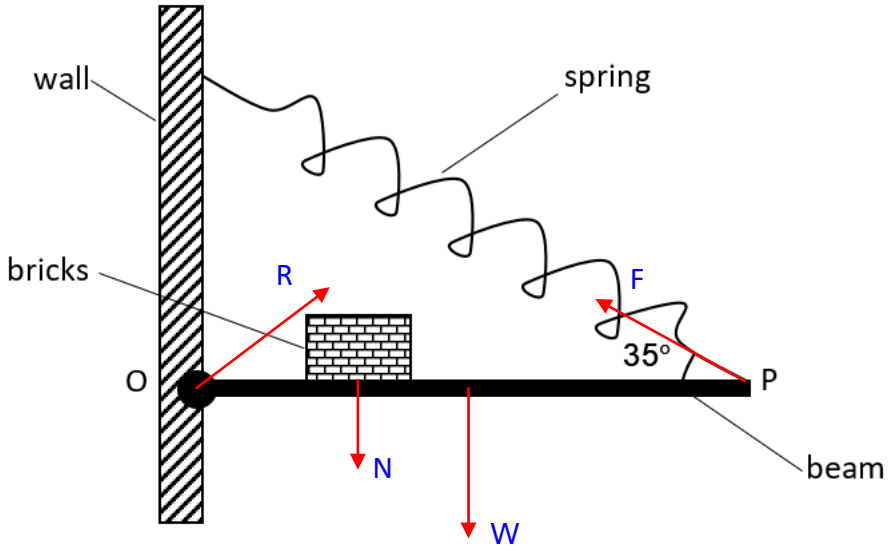
Qns	Answer	Marks
1(a)	Moment of a force is the product of the force and its perpendicular distance to the pivot.	B1
1(b)(i)	$\cos 35° = \frac{6.8}{y}$ $y = \frac{6.8}{\cos 35°} = 8.3$ m extension $x = 8.3 - 7.0$ $= 1.3$ m	M1 A0
1(b)(ii)	 W: weight of beam (or gravitational force by earth on beam) N: normal contact/reaction force by bricks on beam F: force by spring on beam (or spring force / tension / elastic force) (EITHER) R: contact/reaction force by wall/hinge on beam (OR) $R_{vertical}$: vertical component of contact/reaction force by wall/hinge on beam $R_{horizontal}$: horizontal component of contact/reaction force by wall/hinge on beam (Alternatively): normal contact/reaction force by wall/hinge on beam Correct position and direction of all forces Length of arrow of W longer than N	B1 B1

Qns	Answer	Marks
1(b)(iii)	For rotational equilibrium, taking moment about the pivot O, sum of clockwise moment = sum of anticlockwise moment $N(1.8)+W(3.4)=(F\sin 35°)(6.8)$ $(120)(9.81)(1.8)+(190)(9.81)(3.4)=(F\sin 35°)(6.8)$ $F=2168$ N $F=2200$ N or $F=2170$ N $k=\frac{F}{x}=\frac{2200}{1.3}$ or $k=\frac{F}{x}=\frac{2170}{1.3}$ $=1692$ N m^{-1} $=1669$ N m^{-1} $=1700$ N m^{-1} (2 s.f.) $=1700$ N m^{-1} (2 s.f.) Accept both 2 s.f. and 3 s.f. **Note: accept 1670 N m^{-1}, 1690 N m^{-1} and 1700 N m^{-1}.**	**M1** **A1**
1(b)(iv)	$E_P=\frac{1}{2}kx^2$ $=\frac{1}{2}(1700)(1.3)^2$ $=1437$ $=1400$ J (2 s.f.) Note: Can also use $E_P=\frac{1}{2}Fx$ or $E_P=\frac{1}{2}\left(\frac{F^2}{k}\right)$ Accept both 2 s.f. and 3 s.f. Allow ECF. **Note: accept 1400 J, 1410 J, 1430 J and 1440 J**	 **C1** **A1**
1(b)(v)	As the pile of bricks is moved towards the pivot O, the clockwise moment about the pivot (due to the bricks) decreases (as the perpendicular distance from pivot to line of action of the force decreases). (The total clockwise moment decreases). A lower anti-clockwise moment is needed to maintain equilibrium. Therefore, the force exerted by the string (on the beam) decreases.	**B1** **B1**

Qns	Answer	Marks
2**(a)**	g and r are constants so a is proportional to x the negative sign shows that a and x are in opposite directions/ a is always directed towards the equilibrium	**B1** **B1**
2(b)	since $a = -\frac{gx}{r}$ and $a = -\omega^2 x$ so $\omega^2 = \frac{g}{r}$ $\omega^2 = \frac{9.81}{0.28}$ $\omega = 5.9$ rad s^{-1} *(5.92)*	**M1** **A1**
2(c)(i)	Negative sine graph max a = $\omega^2 x_o$ = 5.92 (0.05) = 1.7 m s^{-2}	**B1** **B1**
(c)(ii)	since this is a light damping oscillation, period remain the same as before and remains constant over time Amplitude consistently decreases and first peak must be less than 5.0cm	**B1** **B1**
(c)(iii)	Loss in energy $= \frac{1}{2} m\omega^2 x_{oi}{}^2 - \frac{1}{2} m\omega^2 x_{of}{}^2$ $= \frac{1}{2} \times 0.090 \times 5.9^2 \times 0.050^2 - \frac{1}{2} \times 0.090 \times 5.9^2 \times 0.025^2$ M1 $= 0.294 = 0.29$ J A1	**M1** **A1**

Qns	Answer	Marks
3(a)(i)	$Q_{\text{water at 5.0°C to water at 0°C}} = m_{\text{water}} c_{\text{water}} \Delta T$ $= 220 \times 4.18 \times (5.0 - 0)$ $= 4598$ J $= 4600$ J (2 s.f.)	**B1**
3(a)(ii)	$Q_{\text{water at 0°C to ice at 0°C}} = m_{\text{water}} L_{\text{f}}$ $= 220 \times 334$ $= 73480$ J $= 73500$ J (3 s.f.)	**B1**

Qns	Answer	Marks
	Note: accept 73000 J and 73500 J.	
3(b)	$Q_{\text{ice at -8.0°C to ice at 0°C}} = m_{\text{ice}} c_{\text{ice}} \Delta T$ $= 500 \times 2.09 \times \left[0 - (-8.0)\right]$ $= 8360$ J (3 s.f.) **Note: accept 8360 J and 8400 J.**	**B1**
3(c)	0 °C Since the heat needed to raise the temperature of the original ice to 0 °C (= 8360 J) is greater than the heat released by water as its temperature decreases to 0 °C (= 4598 J), but is less than the total heat released by water as it freezes to 0 °C ice (= 4598 + 73480 = 78078 J), the original water will not fully freeze into 0 °C ice. The final state will be water-ice mixture at 0 °C. **Note:** When the temperature of the water decreases to 0 °C (releasing 4598 J of heat), the temperature of the original ice is still below 0 °C (to raise it to 0 °C, 8360 J of heat is needed). Hence, the original water will continue to partially freeze, until the original ice is at 0 °C. The final product is a water-ice mixture at 0 °C.	**A1** **M1**

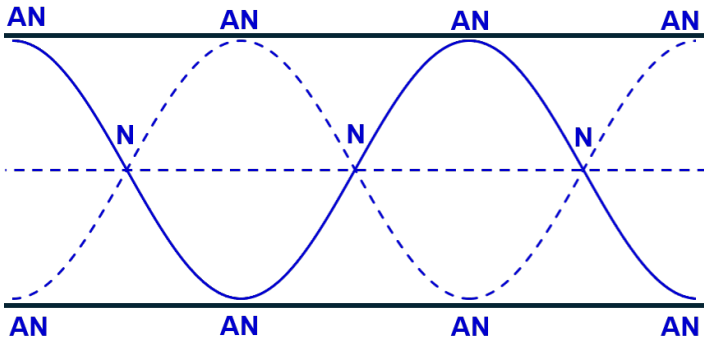
Qns	Answer	Marks
4(a)(i)	Power received = $\frac{\text{Area of source}}{4\pi r^2} \times$ Power of source $= \frac{450 \times 10^{-4}}{4\pi (3500)^2} \times 15.0$ [M1] $= 4.4 \times 10^{-9}$ W [A1]	
4(a)(ii)	Energy from the source radiates uniformly in all directions	**B1**

Qns	Answer	Marks
4(a)(iii)	Intensity $\propto$ Amplitude^2 Intensity = $\frac{\text{Power}}{4\pi r^2} \propto \frac{1}{r^2}$ Hence $A \propto \frac{1}{r}$ $\frac{A_{3.5}}{A_{2.5}} = \frac{r_{2.5}}{r_{3.5}}$ $\frac{A_{3.5}}{A_o} = \frac{2.5}{3.5}$ [M1] $A_{3.5} = 0.71 A_o$ [A1]	
4(b)(i)	Oscillations/ vibrations occur in a single direction (in a plane) that is perpendicular to the direction of transfer of energy/propagation.	**B1**
4(b)(ii)	Apply Malus' Law, $I_1 = I_0(\cos 40°)^2$ $I_2 = I_0(\cos 40°)^2(\cos 80°)^2$ **M1** $= 0.018 Io$ (2 s.f.) **A1**	

Qns	Answer	Marks
5(a)	When an object freefalls, the **ONLY** force acting on the object is its weight (= *mg*). **M1** Hence, the freefall acceleration $a = \frac{F}{m} = \frac{mg}{m} = g$ is equal to the gravitational field strength *g*. **A1** OR By definition of gravitational field strength, $g = F_{\text{graviational}}/m$ and from Newton's 2nd Law, for a freely falling body, the **ONLY** force acting is gravitational force hence $F_{\text{net}} = mg = ma$. Therefore, $a = g$.	
5(b)	$g = \frac{GM}{r^2}$ $M = \frac{gr^2}{G} = \frac{9.81 \times (6.37 \times 10^6)^2}{6.67 \times 10^{-11}} = 5.9679 \times 10^{24} = 5.97 \times 10^{24}$	**B1**

Qns	Answer	Marks
5(c)	$\phi = -\frac{GM}{r} = -\frac{6.67\times10^{-11}\times5.97\times10^{24}}{6.37\times10^{6}} = -6.25\times10^{7}\ \text{J kg}^{-1}$	B1
5(d)	The gravitational potential is taken to be zero at infinity. Since the gravitational force is attractive, the work done by the external force to bring the test mass from infinity to a point is negative. Hence, the gravitational potential is negative.	B1 B1 A0

Qns	Answer	Marks
6(a)	The principle of superposition states that when two or more waves meet at a point, the resultant displacement is the sum of the individual displacements of each wave at that point. **Note:** Do **NOT** accept if students mention the "amplitude add up". The correct term is displacement.	B1
6(b)	Incident and reflected waves are of the same type, frequency and wavelength, and have the same speed, approximately the same amplitude and travel in opposite direction along the same tube. Incident wave from the loudspeaker reflects at the end / right side of the glass tube such that they superpose to form a stationary wave. **Note:** Do not accept if students write meet / overlap without superpose. The word interfere is not accepted as well.	B1 B1 B1
6(c)(i)	Amplitude of stationary wave: 0.06 + 0.06 = 0.12 mm	B1

Qns	Answer	Marks
6(c)(ii)	 General shape (include spacing) of a stationary wave (including dotted cosine curve – no solid line) with displacement antinodes at the ends of the glass tube. Stationary waves fulfilling the 3rd harmonic, n = 3, since there are three pressure antinodes (three displacement nodes) *#Although labelling **N** and **AN** is not a marking point, it is good practice to write it down.* **Note:** Students should know that a pressure antinode corresponds to a displacement node. Hence, there should be 3 displacement nodes which correspond to the 3rd harmonic for a pipe with two open ends.	**B1** **B1**
6(c)(iii)	$L = 1.5\lambda \Rightarrow \lambda = \frac{L}{1.5}$ $v = f\lambda \Rightarrow f = \frac{v}{\lambda} = \frac{v}{\left(\frac{L}{1.5}\right)} = \frac{320}{\left(\frac{15.0}{1.5}\right)} = 32 \text{ Hz}$	**M0** **A1**
7(a)(i)	${}^{210}_{83}Bi \rightarrow {}^{0}_{-1}e + {}^{210}_{84}Po$ + (anti) neutrino Accept beta particle	**B2**
(a)(ii)	total momentum/ total mass-energy/ total nucleon no/ total charge. Any two.	**A2**
(b)(i)	Mass of protons + neutrons = [(83 x 1.00729) + (127 x 1.00867)] u = 211.706 u Mass defect = 211.706 u - 209.939 u Mass defect = 1.767 u	**M1** **M1** **A0**

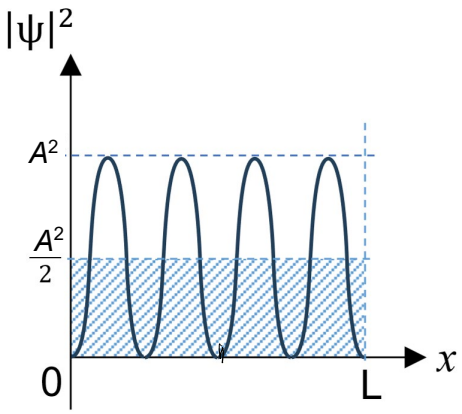
Qns	Answer	Marks
(b)(ii)	Binding energy = $1.767 \times 1.66 \times 10^{-27} \times (3.00 \times 10^{8})^2$ $= 2.64 \times 10^{-10}$ J Binding energy per nucleon = $[2.64 \times 10^{-10} / 1.60 \times 10^{-19} \times 10^{6}]/210$ = 7.86 MeV	M1 A1

Qns	Answer	Marks
8(a)1	Since the circular path is in the clockwise direction it suggests that the magnetic force must point rightwards (from left to right) Using Fleming's Left Hand Rule, (when the magnetic field is out of the paper, and the magnetic force is towards the right), this is only possible if the particle is positively charged.	B1 B1
8(a)2	For the positively charged particle to accelerate upwards it must experience an electric force that acts in the upward direction. This occurs only if the electric field points upwards which corresponds to a higher electric potential at the bottom plate.	B1 B1
8(b)(i)	The electric field between parallel plates is uniform. $E = \frac{\Delta V}{\Delta d} = \frac{V_0}{d} = \frac{V}{0.5d}$ Hence, $V = \frac{V_0}{2}$	M1 A0
8(b)(ii)	Gain in kinetic energy = Loss in electric potential energy $\frac{1}{2}mv^2 - 0 = q\Delta V = \frac{qV_0}{2}$ $\therefore v^2 = \frac{qV_0}{m}$ $v = \sqrt{\frac{qV_0}{m}}$	M1 A0
8(c)	The magnetic force always acts perpendicular to the direction of velocity (infinitesimal tangential displacement along the charged particle's path). No work is done, its kinetic energy remains constant and thus, the speed does not increase.	B1 B1

Qns	Answer	Marks
8(d)	The magnetic force (F_B) provides for the centripetal force (F_c) $F_B = F_c$ $Bqv = mr\omega^2 \Rightarrow Bq(r\omega) = mr\omega^2$ $\therefore \omega = \frac{Bq}{m}$ $\omega = \frac{2\pi}{T}$ $\frac{2\pi}{T} = \frac{Bq}{m}$ $\therefore T = \frac{2\pi m}{Bq}$ Since the particle travels half a circle, the time travelled in the upper dee is: $t = \frac{T}{2}$ $= \frac{\pi m}{Bq}$ **Alternative solution:** The magnetic force (F_B) provides for the centripetal force (F_c) $F_B = F_c$ $Bqv = \frac{mv^2}{r} \Rightarrow v = \frac{Bpr}{m}$ Since $v = \frac{\pi r}{t}$, where πr is the distance travelled and t is the time taken for the semicircular path $\frac{\pi r}{t} = \frac{Bqr}{m}$ $\therefore t = \frac{\pi m}{Bq}$	B1 C1 M1 M1 A0 B1 C1 M1 M1 A0
8(e)	The time taken is independent of the velocity of the particle (only depends on m, B and q), Hence, it will always be the same as that found in (d).	M1 A1
8(f)	t = 1.6 ns $B = \frac{\pi m}{tq} = \frac{\pi}{(1.6\times10^{-9})(1.0\times10^{8})} = 19.63495 = 20$ T (2.s.f)	C1 A1
8(g)	At the upper dee, the charged particle has a kinetic energy of $\frac{qV_0}{2}$ Gain in kinetic energy = Loss in electric potential energy = qV_0	 C1

Qns	Answer	Marks
	Thus, total kinetic energy = $\frac{qV_0}{2} + qV_0 = 1.5qV_0$	**A1**
8(h)	When the particle first travels into the upper dee, it has a kinetic energy of 0.5 qV_0. The total kinetic energy of the particle can then be written as E_K = (0.5 + n) qV_0, where n is the number of times the particle is in either of the dees after exiting the upper dee for the first time. Since the kinetic energy is 6.5 qV_0, n = 6. By doing a manual count, the particle will exit from the upper dee and the it passes the lower dee is a total of 3 times. **#Food for thought:** To generalize this: If n = odd, then we can tell that the particle exits from the lower dee and the number of times it will pass the lower dee is $\frac{(n+1)}{2}$ If n = even, then we can tell that the particle exits from the upper dee and the number of times it will pass the lower dee is $\frac{n}{2}$	**C1** **A1**

Qns	Answer	Marks
9(a)(i)	The square of the wavefunction amplitude $\lvert\psi(x)\rvert^2$ is the probability density function of the particle at position x.	**B1**
9(a)(ii)	The probability of the particle residing between x_1 and x_2 is given by the area under the $\lvert\psi(x)\rvert^2$-x graph between x_1 and x_2.	**B1**
9(b)(i)	n = 4.	**B1**

Qns	Answer	Marks
9(b)(ii)	 area under $= \frac{A^2}{2}L = 1$ $A = \sqrt{\frac{2}{L}}$	B1 M1 A1
9(b)(iii)	By symmetry of the graph, the area under the $\|\psi(x)\|^2$-x graph between $x = \frac{L}{8}$ and $x = \frac{L}{2}$ is $\frac{3}{8}$ of the total area, which is equal to 1. Hence, the probability is $\frac{3}{8} = 0.375$.	B1 B1
9(b)(iv)	$E = \frac{h^2}{8mL^2}n^2 = \frac{(6.63\times10^{-34})^2}{8\times9.11\times10^{-31}\times(3.5\times10^{-10})^2}\times4^2 = 7.87778\times10^{-18}$ J $= \frac{7.87778\times10^{-18}}{1.60\times10^{-19}} = 49.2$ eV	C1 A1
9(c)(i)1	absorption spectrum dark lines on colourful/continuous background	B1 B1
9(c)(i)2	When white light containing all frequencies passes through the cold gas, some photons whose energies are exactly equal to the differences between the atomic energy levels will be absorbed. Since the energy levels are discrete, only photons of certain frequencies are absorbed. Photons of these frequencies will be re-radiated but in all directions when the atoms deexcite.	B1 B1 B1

Qns	Answer	Marks
	Consequently, the parts of the spectrum corresponding to these frequencies appear darker in comparison to the other frequencies.	**B1**
9(c)(ii)	emission spectrum colourful lines on dark background	**B1**
9(c)(iii)	$E_{\text{photon}} = -3.4 - (-13.6) = 10.2 \text{ eV} = 10.2 \times 1.6 \times 10^{-19} = 1.63 \times 10^{-18} \text{ J}$ $E_{\text{photon}} = \frac{hc}{\lambda} = \frac{6.63 \times 10^{-34} \times 3.0 \times 10^{8}}{\lambda} = 1.63 \times 10^{-18} \text{ J}$ $\lambda = \frac{6.63 \times 10^{-34} \times 3.0 \times 10^{8}}{1.63 \times 10^{-18}} = 1.22 \times 10^{-7} = 122 \text{ nm}$ in the ultraviolet region of the spectrum	**M1** **A1** **A1**